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Serogroup W invasive meningococcal disease associated with Hajj/Umrah pilgrimage and travel to the Middle East and Asia in 2024 and 2025

Background

Invasive meningococcal disease (IMD) due to bacterial infection with Neisseria meningitidis typically presents as meningitis, septicaemia, or a combination of the two. IMD is associated with significant morbidity and mortality, with survivors often experiencing lifelong complications. There are 12 known meningococcal serogroups, determined by the composition of the outer polysaccharide capsule, with most disease caused by serogroups A, B, C, W, X and Y.

The highly successful teenage MenACWY vaccination programme, introduced in the UK in 2015, led to large reductions in IMD due to the targeted capsular groups across all ages through a combination of direct and indirect (herd) protection (1). The COVID-19 pandemic and implementation of social distancing measures and lockdown periods across the UK in 2020 and 2021 had a marked impact on the incidence of IMD (2), which further accelerated population control of MenC, MenW and MenY (3). Following the withdrawal of all COVID-19 control measures from July 2021, IMD incidence increased, but mainly due to MenB cases, which have predominated in England and Wales, with only a few cases caused by the other serogroups.

In April 2024, the UKHSA Meningococcal Reference Unit (MRU) confirmed 3 MenW cases in England and Wales amongst Umrah pilgrims within days of their return from Saudi Arabia. In each case, there was no record of MenACWY vaccination before travel, despite this being a Hajj/ Umrah visa requirement (4). These cases were part of an international outbreak of 12 IMD cases confirmed in the UK, France and the USA – in children (2 cases) and adults (4 aged 25-44 years, 4 aged 45-64 years, 2 aged 65+ years) – associated with recent travel to Saudi Arabia for Umrah (5). The 10 adult cases had visited Saudi Arabia themselves, and the 2 children were contacts of returned pilgrims. Of the 11 cases with isolates available for testing, 10 were confirmed as MenW of the hyper-invasive sequence type 11 clonal complex (MenW:cc11). The other was an unrelated MenC strain.

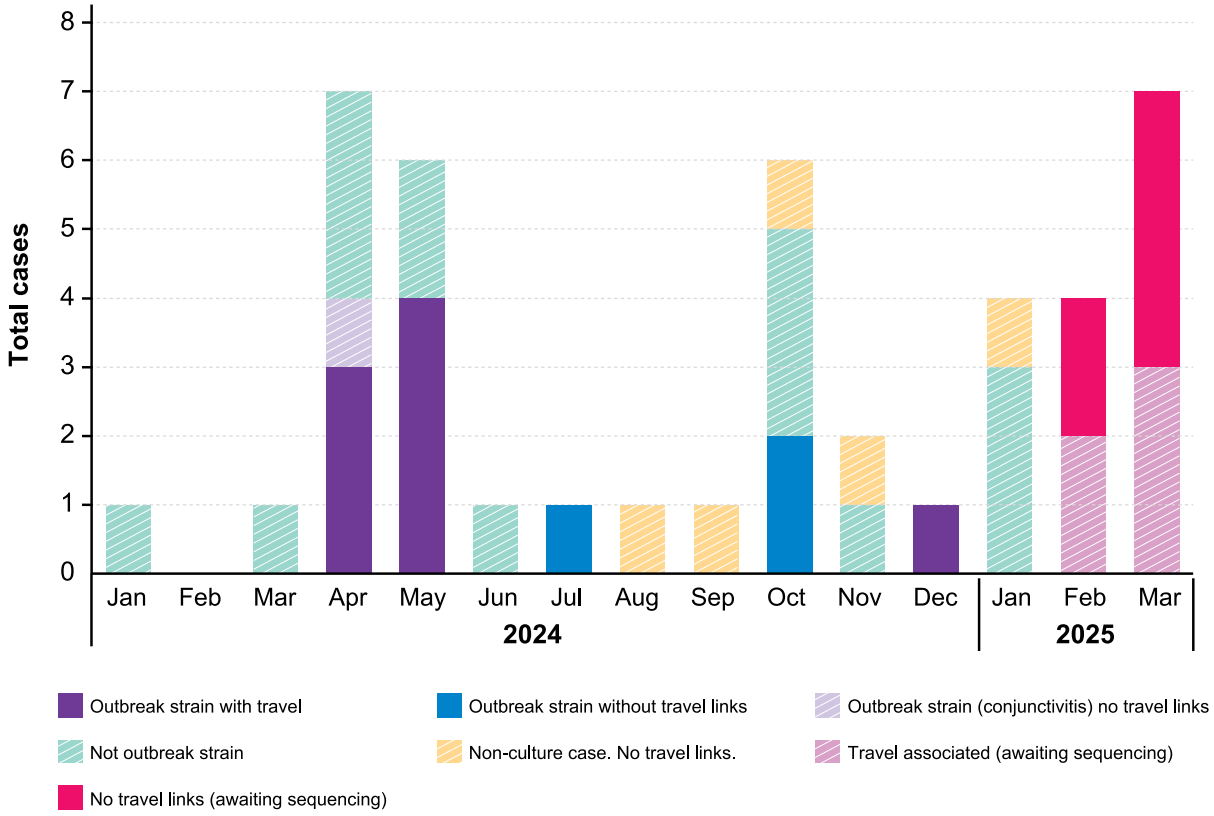
Current epidemiology

Between January 2024 and March 2025, a total of 42 cases of MenW IMD in England and Wales were confirmed by the MRU, of which:

- 9 were associated with recent travel to Saudi Arabia and
- 4 were recent travellers to other countries in the Middle East or Asia (see figure)

Of the 26 IMD cases with cultures sequenced between January 2024 and January 2025, all 8 travel-related cases that arose in that period were confirmed as the MenW cc11 outbreak strain. An additional 3 MenW IMD cases and 1 meningococcal conjunctivitis case confirmed in that period have been microbiologically linked to this outbreak strain. These arose in cases with no recent travel and no known links to recent travellers in three geographically distinct areas, suggesting some community transmission may have occurred. Sequencing is outstanding for February-March 2025, including 5 travel-associated cases. All MenW cases were sensitive to the tested antibiotics (penicillin, rifampicin, ciprofloxacin and ceftriaxone).

Confirmed cases of invasive meningococcal disease serogroup W* (n=42) in England and Wales, January 2024 to March 2025 (*and 1 outbreak strain conjunctivitis case)



Source. UKHSA Meningococcal reference Unit.

Discussion

The Hajj is an annual Islamic pilgrimage to Mecca in Saudi Arabia during specified dates (4 to 9 June in 2025), whilst Umrah is a pilgrimage to Mecca that can be performed at any time. Millions of international travellers perform the Hajj and Umrah pilgrimages each year, many during the month of Ramadan. IMD outbreaks associated with Hajj led to the 2002 introduction of a requirement for documented MenACWY vaccination for pilgrims. MenACWY vaccination is now required for Umrah and Hajj pilgrims (4). within five years of prior MenACWY conjugate vaccination, or within three years of polysaccharide vaccination, and administered at least 10 days before arrival. But enforcement is challenging for Umrah, in particular, as pilgrims can travel to Saudi Arabia on a different visa (such as a tourist visa) before performing Umrah (6).

All pilgrims travelling to Saudi Arabia should be appropriately immunised with the MenACWY vaccine to help protect themselves and their families and communities on return from travel. Every case of IMD and meningococcal infection of the conjunctiva should be reported to UKHSA Health Protection Teams, to allow appropriate public health action to be undertaken as a matter of urgency. This includes assessment of meningococcal vaccination status and travel history in the previous month, for the case as well as their close contacts. Rifampicin is recommended for first-line prophylaxis in close contacts of both IMD and conjunctivitis cases with recent travel to the Middle East or Asia. Rifampicin should also be offered for such cases not treated with cephalosporins (7). All young people who did not receive their teenage MenACWY conjugate vaccine at 13-15 years of age remain eligible for vaccination through their School Age Immunisation Service, where appropriate, or GP practice, until their 25th birthday.

The UKHSA Vaccination and Immunisation Division and Meningococcal Reference Unit will continue to closely monitor the situation. The Ministry of Hajj and Umrah in Saudi Arabia has set 29 April as the deadline for foreign Umrah pilgrims to leave the country so that preparations for Hajj season can begin; as a result, there may be a temporary fall in cases (8).

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